A Vision Transformer (ViT)-based Domain Adaptation Framework for Wildlife Camera-Trap Species Classification

ACM40960: Project in Math Modelling | Xuan Bach Nguyen (25206963) - University College Dublin

1. Problem Formulation

Camera traps are widely used in biodiversity monitoring, and the volume of accumulated images has motivated the application of deep learning techniques to automatically identify wildlife. However, night-vision captures are often monochrome infrared images, and quality can vary because of darkness, motion blur, occlusion, and sensor settings, which makes species identification harder than in daylight imagery.

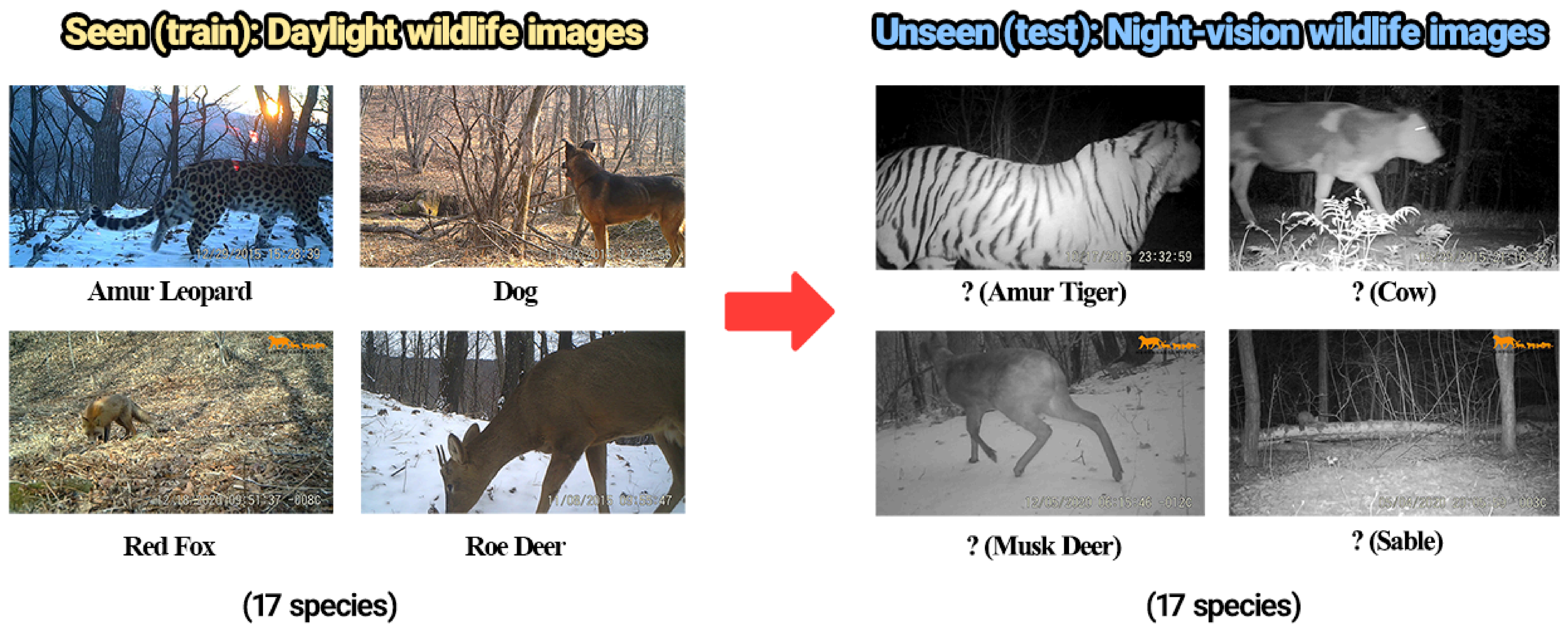

Figure 1. Problem formulation. From labelled daytime wildlife images, the model classifies species in camera-trap nighttime images.

The proposed framework leverages **Vision Transformer (ViT)** with data adaptation and deep learning techniques to bridge the domain gap between daytime and nighttime images. This problem is best framed as a **domain generalisation** task, where the model identifies the species captured by camera-trap using only daytime images for training.

2. Dataset Introduction

Dataset: NTLNP image dataset for object detection of wildlife. The full daytime dataset is used for training, while the nighttime dataset is divided into **20/80 for validation and testing** respectively. **Class-balanced sampling** is applied to ensure all species appear in both sets and avoid bias during validation.

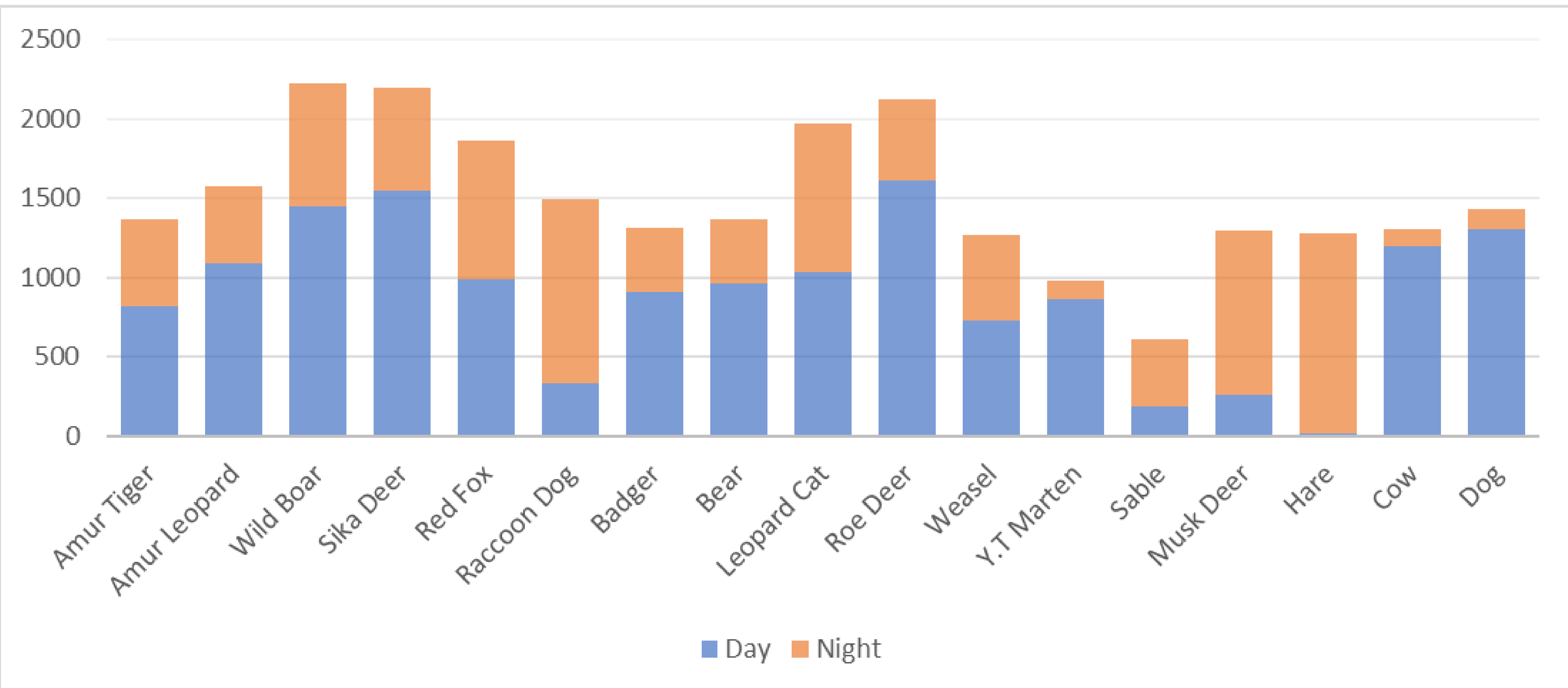

Figure 2. NTLNP dataset statistics

Main challenges:

- **Severe class imbalance:** some species have more than 1,000 daytime images, in contrast to ones with less than 200.
- **Domain shift:** model performance drops under drastic changes in environment.
- **Test image quality:** monochrome, motion blur, occlusion images.

3. Backbone Architecture

This framework uses **DINOv2-base** as the backbone model for feature extraction since self-supervised ViT features have shown strong semantic structure and good transfer behaviour, making them suitable for representation learning in difficult visual conditions.

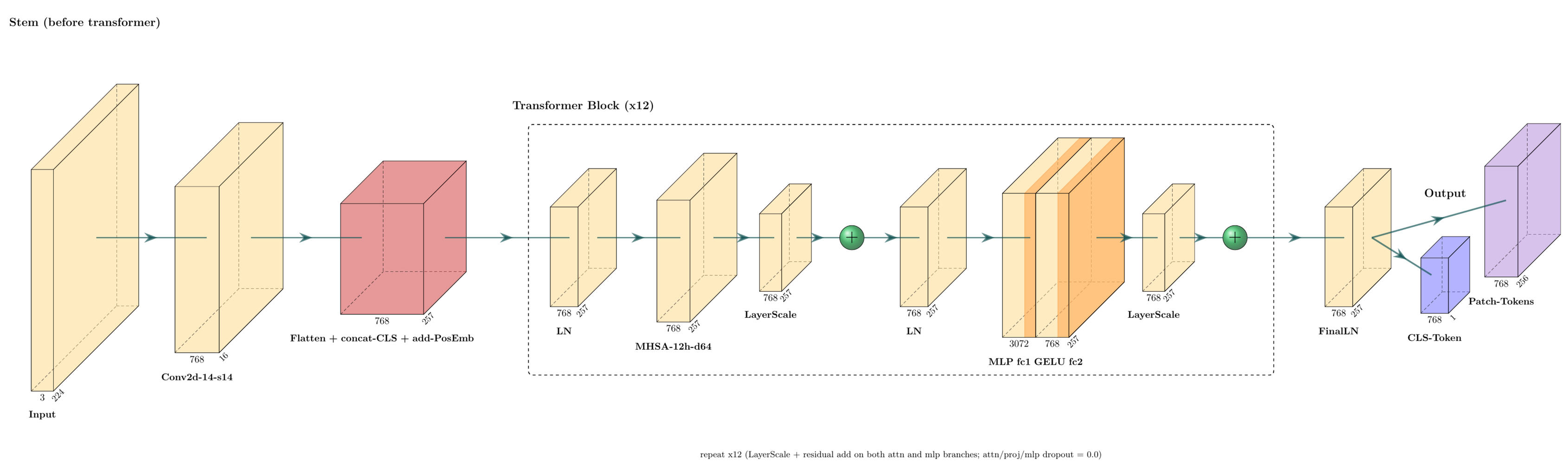

Figure 3. DINOv2-base architecture.

The input image is segmented into **14x14 patches with positional embeddings** and a **learnable [CLS] embedding** prepended to the sequence to capture global image information, then passed through **12 Transformer encoder blocks**, resulting in a final **CLS token** as image representation and **patch tokens** with preserved spatial relationship.

References

[1] Oquab, Maxime, et al. "Dinov2: Learning robust visual features without supervision." arXiv preprint arXiv:2304.07193 (2023).
[2] Khosla, Prannay, et al. "Supervised contrastive learning." Advances in neural information processing systems 33 (2020): 18661-18673.
[3] Li, Yanfeng, et al. "Research on target image classification in low-light night vision." Entropy 26.10 (2024): 882.
[4] Tan, Mengyu, et al. "Animal detection and classification from camera trap images using different mainstream object detection architectures." Animals 12.15 (2022): 1976.

4. Framework Workflow

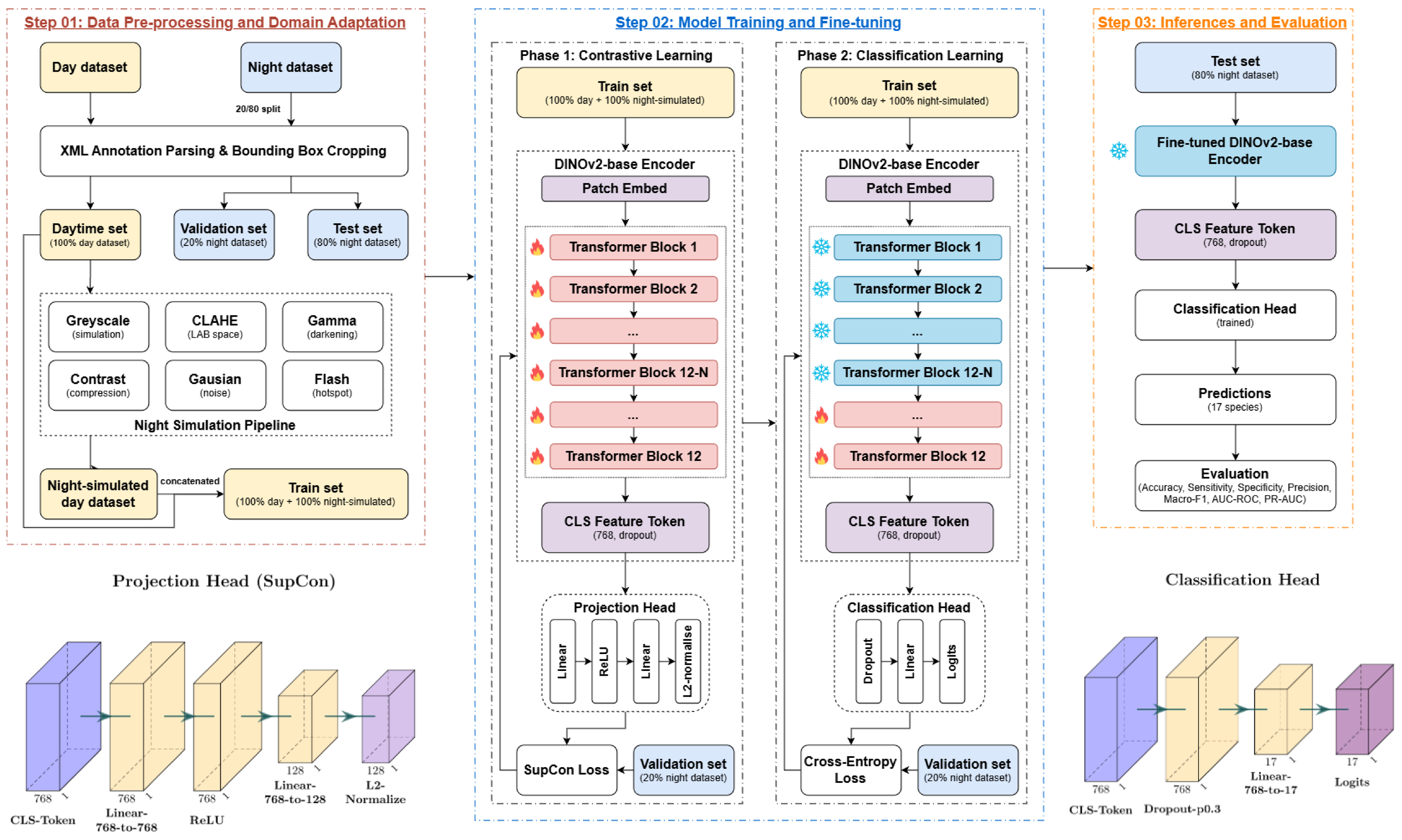

Figure 4. Workflow. The framework consists of 03 stages.

- The daytime dataset is simulated to match the **night environment** in the first stage.
- During each phase in the second stage, the **first 5 epochs** are spared to shape the head with **all transformer blocks frozen**.
- The **frozen fine-tuned DINOv2-base encoder** and the **classification head** after the second stage is used for inference and evaluation in the last stage.

5. Results

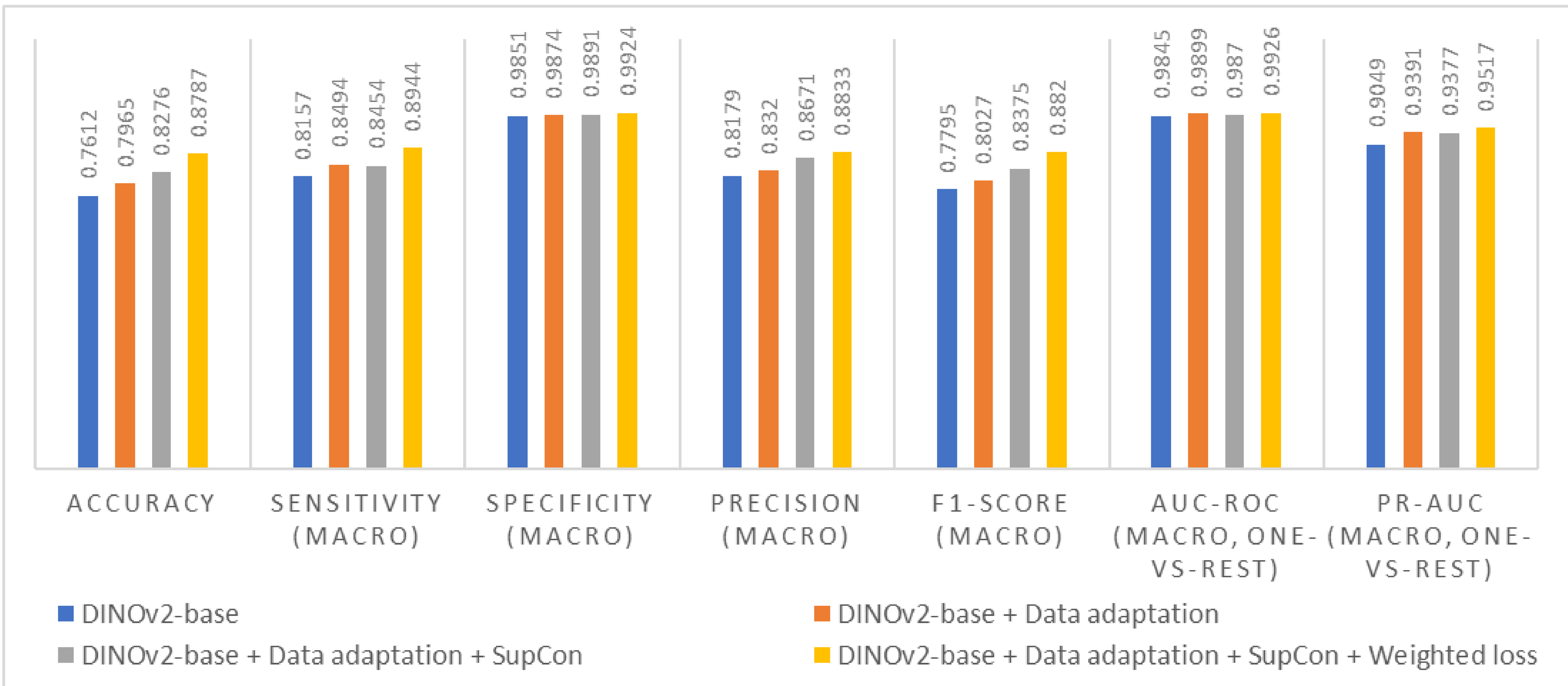

Figure 5. Model performance comparison. The full model achieves the best results across all metrics.

Model performance: Overall, every component contributes meaningfully to the framework, and the full model outperforms the baseline model by **roughly 8-15%** depending on the evaluation metrics except specificity and AUC-ROC, which are already close to 1.

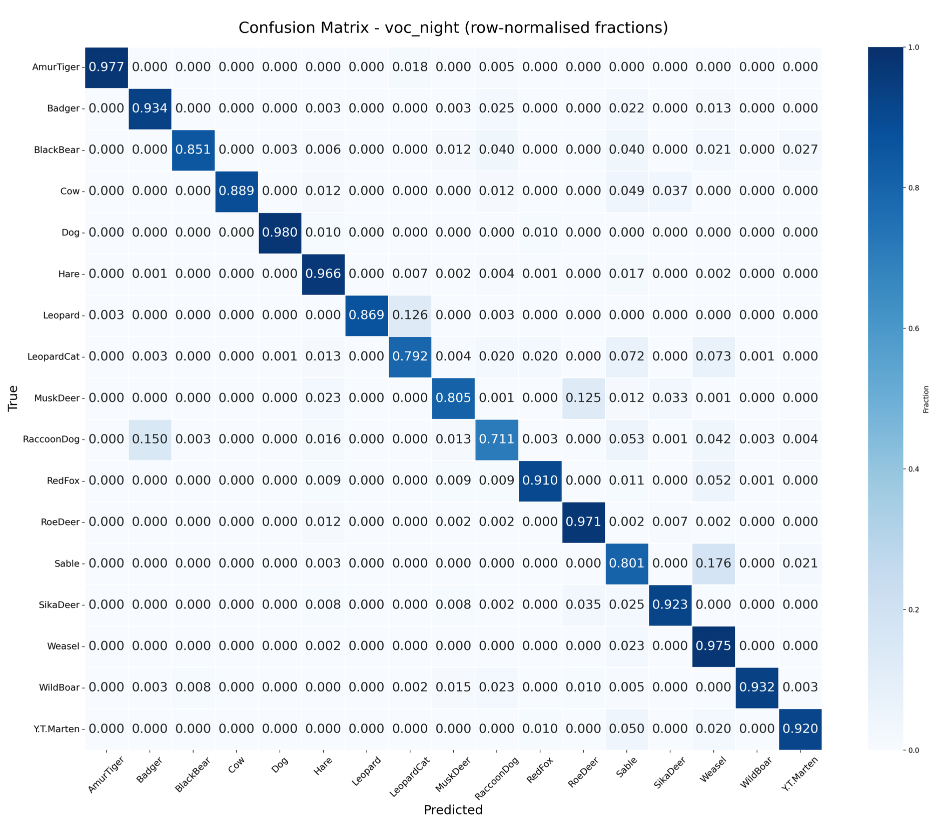

Figure 6. Per-species confusion matrix

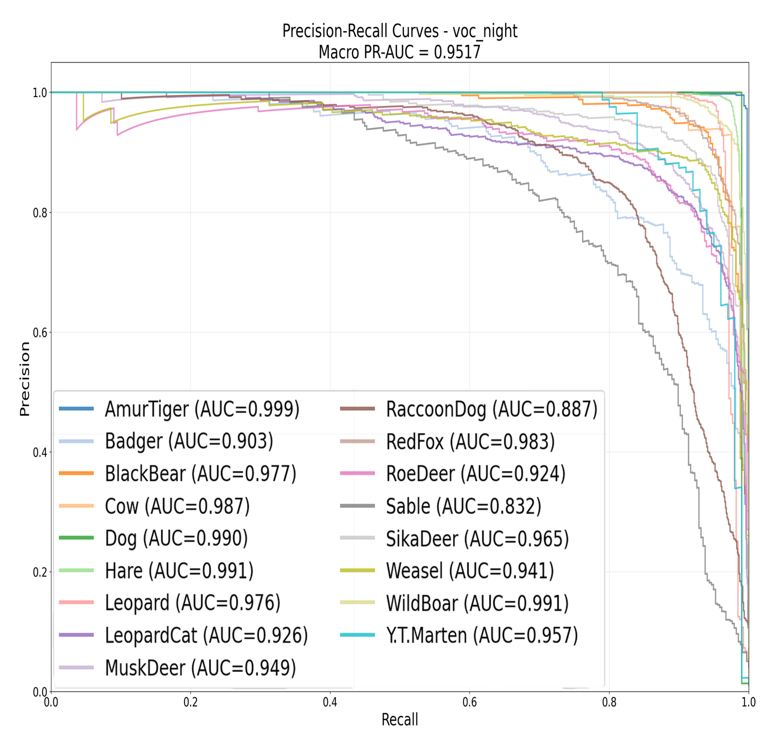

Figure 7. Per-species Precision-Recall curve

Per-species analysis:

- **5/17 species have $\geq$ 96% classification accuracy**, with another **5 have** $\geq$ **90%** and another **5 have** $\geq$ **80%**. Leopard cat and raccoon dog perform worst.
- **13/17 species have AUC $\geq$ 0.92**. Sable, raccoon dog, and badger decay steepest.
- Common misclassification: Sable $\rightarrow$ Weasel, RaccoonDog $\rightarrow$ Badger, Leopard $\rightarrow$ LeopardCat, MuskDeer $\rightarrow$ RoeDeer.

6. Conclusion

- **All components** implemented enhances the classification accuracy meaningfully, achieving **slightly under 90%** in accuracy, sensitivity, precision, and F1-score for the full model. Domain generalisation task succeeds broadly at the species level.
- **Data imbalance** is addressed and partially but not completely resolved. Small nocturnal animals are harder to distinguished.
- Future developments may include **self-supervised pre-training** on unlabelled wildlife images, adding **test-time adaptation**, and **sequence-level modelling**.